EXAM POV NOTES (Q&A FORMAT)

## Topic 1: Linear Regression

### Q1 Explain Linear Regression in Machine Learning, including its definition, types, steps, an example, advantages, and disadvantages. (10 Marks / 11 points)

Linear regression is a statistical method used in machine learning to **model the relationship between a dependent variable (target) and one or more independent variables (features)** by fitting a linear equation to observed data.

**Types of Linear Regression**:
- ◦ **Simple Linear Regression**: Involves **one independent variable**.
- ◦ **Multiple Linear Regression**: Involves **multiple independent variables**.

**Steps in Linear Regression**:
- ❖ **Data Collection**: Gather a dataset with input features and corresponding outputs.
- ❖ **Data Preprocessing**:
  - ▪ Handle missing values.
  - ▪ Normalize or standardize data if necessary.
  - ▪ Split data into training and testing sets.
- ❖ **Model Training**:
  - ▪ Fit the linear regression model using the training data.
  - ▪ Calculate the coefficients, typically using **Ordinary Least Squares (OLS)**, to find the best-fitting line or hyperplane by minimizing the difference between actual and predicted values.
- ❖ **Prediction**: Use the trained model to predict output values for new data.
- ❖ **Evaluation**: Measure the model's performance using metrics such as Mean Squared Error (MSE) or **R-squared**.

• **Example: Predicting House Prices**:
- **Problem**: Predict the price of a house based on its size (in square feet) and the number of bedrooms.
- **Data Preparation**: Features would be 'Size (sq. ft.)' and 'Bedrooms', and the target would be 'Price'.
- **Model Training**: The model learns coefficients (weights) for the features. For example, a trained model might use an equation like $\text{Predicted Price} = \theta_0 + \theta_1 \times \text{Size} + \theta_2 \times \text{Bedrooms}$.
- **Prediction**: For a new house (e.g., 2500 sq. ft., 4 bedrooms), the model would predict its price using the learned equation.
- **Evaluation**: The model's accuracy is calculated using R-squared or MSE on the test dataset. An R-squared value of 0.85 indicates that 85% of the variance in house prices is explained by the model.

• **Advantages of Linear Regression**:
- ◦ **Simple to implement and interpret**.
- ◦ **Computationally efficient**.
- ◦ Works well when the relationship between features and the target is linear.

• **Disadvantages of Linear Regression**:
- ◦ Assumes a linear relationship, which may not always be the case.
- ◦ Sensitive to outliers.
- ◦ Can overfit in the presence of too many features.

### Q2 Outline the key steps involved in a Linear Regression model from data collection to evaluation. (5 Marks / 6 points)

- The key steps involved in a Linear Regression model are:
- **Data Collection**: This initial step involves gathering a dataset that includes both the input features (independent variables) and their corresponding output values (dependent variable or target).
- **Data Preprocessing**: Before training, the data needs preparation. This includes handling any missing values, normalising or standardising data if required, and **splitting the dataset into training and testing sets**.
- **Model Training**: In this phase, the linear regression model is fitted using the training data. The core task is to **calculate the coefficients** for the linear equation, typically by employing the **Ordinary Least Squares (OLS) method** to minimise the difference between actual and predicted values.
- **Prediction**: Once the model is trained, it can be used to **predict output values for new, unseen data** based on the learned linear relationship.
- **Evaluation**: The final step involves measuring the model's performance. This is done using various metrics, such as **Mean Squared Error (MSE)** or **R-squared**, to assess how well the model's predictions align with actual values.

### Q3 Define Linear Regression and list its two main types. (2 Marks / 2 points)

**Linear Regression** is a statistical method used in machine learning to **model the relationship between a dependent variable (target) and one or more independent variables (features)** by fitting a linear equation to observed data.
Its two main types are:

- **Simple Linear Regression**: Involves one independent variable.
- **Multiple Linear Regression**: Involves multiple independent variables.

## Topic 2: Ordinary Least Squares (OLS)

### Q4 Describe Ordinary Least Squares (OLS), its purpose in linear regression, and its key assumptions. (5 Marks / 6 points)

**Ordinary Least Squares (OLS)** is a statistical method used in linear regression to estimate the relationship between a dependent variable and one or more independent variables.
Its primary purpose is to **find the best-fitting line** through the data points by **minimising the sum of the squares of the residuals** (the differences between observed and predicted values). This allows the calculation of coefficients for the linear regression model.
**Key Assumptions of OLS**:

- **Linearity**: The relationship between the dependent and independent variables is linear.
- **Independence**: The residuals (errors) are independent of each other.
- **Homoscedasticity**: The variance of the residuals is constant across all levels of the independent variables.
- **No Multi-collinearity**: Independent variables are not highly correlated with each other.
- **Normality**: Residuals are normally distributed (important for hypothesis testing).

### Q5 (2 Marks / 2 points): What is the primary purpose of Ordinary Least Squares (OLS) in linear regression?

The primary purpose of **Ordinary Least Squares (OLS)** in linear regression is to **find the best-fitting line** through the data points by **minimising the sum of the squares of the residuals** (the differences between observed and predicted values).

## Topic 3: Mean Absolute Error (MAE)

### Q6 Explain Mean Absolute Error (MAE) as a model evaluation metric, including its purpose, characteristics, and advantages. (5 Marks / 6 points)

**Mean Absolute Error (MAE)** is a measure of the **average magnitude of errors in a set of predictions**, without considering their direction. It quantifies the accuracy of a predictive model by **calculating the average absolute difference between observed and predicted values**. It achieves this by summing up the absolute errors and averaging them across all observations.

**Characteristics of MAE**:

- **Scale**: MAE is measured in the **same units as the data**, making it directly interpretable.
- **Sensitivity**: It **treats all errors equally**, providing a straightforward measure of average prediction error.
- **Robustness**: MAE is **less sensitive to outliers** compared to metrics like Mean Squared Error (MSE), as it does not square the errors.

**Advantages of MAE**:

- **Simplicity**: Easy to understand and interpret as it represents the average error magnitude.
- **Direct Interpretability**: Being in the same units as the target variable, it provides a clear measure of error magnitude.
- **Robust to Outliers**: It is less affected by outliers because it doesn't square the error terms.

### Q7 What is Mean Absolute Error (MAE) and what does a lower MAE value signify? (2 Marks / 2 points)

**Mean Absolute Error (MAE)** is a measure of the **average magnitude of errors in a set of predictions**, quantifying the accuracy of a predictive model by calculating the average absolute difference between observed and predicted values.

A **lower MAE indicates better model performance**, with smaller average errors.

## Topic 4: Root Mean Squared Error (RMSE)

### Q8 Discuss Root Mean Squared Error (RMSE) as an evaluation metric, detailing its purpose, characteristics, and how it penalizes errors. (5 Marks / 6 points)

**Root Mean Squared Error (RMSE)** is a standard way to measure the error of a model in predicting quantitative data. It represents the **square root of the average of the squared differences between predicted and observed values**.

Its purpose is to quantify the difference between values predicted by a model and the values observed in reality, giving a sense of how far off predictions are on average. The calculation involves:

1. Calculating the difference between each predicted and actual value.
2. Squaring these differences to remove negative signs and **emphasize larger errors**.
3. Finding the mean of these squared differences.
4. Taking the square root to bring the error back to the original unit of measurement.

**Characteristics of RMSE**:

- **Scale**: RMSE has the **same units as the dependent variable**, allowing direct interpretation within the data's context.
- **Sensitivity**: RMSE is **more sensitive to larger errors** compared to metrics like MAE due to the squaring of differences.
- **Magnitude**: It **gives a higher weight to large errors**, which is useful if significant deviations are particularly undesirable.
- **Penalization of Errors**: RMSE penalises large errors more significantly than small errors because it squares the differences before averaging. This means that a few large errors will have a greater impact on RMSE than many small errors, making it useful in contexts where larger deviations are problematic.

### Q9 Define Root Mean Squared Error (RMSE) and state one of its key characteristics regarding sensitivity to errors. (2 Marks / 2 points)

**Root Mean Squared Error (RMSE)** is a standard way to measure the error of a model in predicting quantitative data, representing the **square root of the average of the squared differences between predicted and observed values**.
One key characteristic is that **RMSE is more sensitive to larger errors** compared to metrics like Mean Absolute Error (MAE) due due to the squaring of differences.

### Topic 5: R-squared (R^2)

### Q10 (5 Marks / 6 points): Describe R-squared (R^2) as a statistical measure, explaining its purpose, interpretation, and limitations.

**R-squared (R^2)**, also known as the coefficient of determination, is a statistical measure that **represents the proportion of the variance in the dependent variable that is predictable from the independent variables**.Its purpose is to **assess the goodness of fit of a regression model**, indicating how well the independent variables explain the variability of the dependent variable. It measures how much of the total variation in the dependent variable is explained by the independent variables in the model.

**Interpretation**:

- **Scale**: R-squared ranges from **0 to 1**.
  A value of **R^2 close to 1 indicates a good fit**, meaning the model explains almost all variability.
- A value close to **0 indicates a poor fit**, suggesting the model does not explain any variability.
- For example, an R^2 of 0.75 means **75% of the variability in the dependent variable is explained by the model**.

**Limitations of R-squared**:

- **Not Always Indicative of Good Fit**: A high R^2 does not necessarily mean the model is good, especially if the model is **overfitting the data**.
- **Doesn't Indicate Causation**: R^2 only indicates correlation, not causation between variables.
- **Affected by the Number of Predictors**: R^2 **will never decrease when more predictors are added** to the model, which can be misleading. Adjusted R^2 is often used to account for this.
- **Sensitivity to Outliers**: Outliers can heavily influence R^2, potentially making the model appear to fit better than it actually does.

### Q11 What is R-squared (R^2) and what does a value close to 1 indicate? (2 Marks / 2 points)

**R-squared (R^2)** is a statistical measure that represents the **proportion of the variance in the dependent variable that is predictable from the independent variables**.
A value of R^2 close to 1 indicates a **good fit**, meaning almost all data points lie exactly on the regression line, and the model explains nearly all the variability in the dependent variable.

### Q12: Explain Bayes' Theorem and elaborate on the Naïve Bayes algorithm, including its foundation, working principle, advantages, disadvantages, and applications.

**Bayes' Theorem** (also known as Bayes' Rule or Bayes' Law) is a fundamental concept used to determine the **conditional probability** of an event A when another event B has already occurred. It allows for determining the probability of a hypothesis using prior knowledge and is dependent on conditional probability.
The **formula** for Bayes' Theorem is **P(A|B) = P(B|A)P(A) / P(B)**.
The key terms involved are:

- **P(A): Prior Probability** – The initial probability of hypothesis A before any evidence is observed.
- **P(B): Marginal Probability** – The probability of event B (the evidence).
- **P(A|B): Posterior Probability** – The probability of hypothesis A given that event B has been observed. This represents the updated probability of A after considering B.
- **P(B|A): Likelihood Probability** – The probability of observing the evidence B given that the hypothesis A is true.

The **Naïve Bayes algorithm** is a **supervised learning algorithm** primarily used for **classification problems**. It operates as a probabilistic classifier, making predictions based on the probability of an object belonging to a particular class.
Its **foundation** lies in Bayes' Theorem, leveraging the calculation of conditional probabilities. The "Naïve" aspect stems from a crucial **assumption**: all features are **independent of each other** given the class variable. This assumption simplifies calculations but can be a limitation in real-world scenarios.

The **working principle** can be illustrated with an example of predicting whether a player should play based on weather conditions:

1. **Calculate Prior Probabilities:** Determine the initial probability of each class (e.g., P(Yes) = 0.71, P(No) = 0.29) from the dataset.
2. **Calculate Likelihoods:** Determine the probability of each feature given the class (e.g., P(Sunny|Yes) = 0.3, P(Sunny|No) = 0.5).
3. **Apply Bayes' Theorem:** Use the formula to calculate the posterior probabilities for each class (e.g., P(Yes|Sunny) = 0.60, P(No|Sunny) = 0.41).
4. **Prediction:** The class with the higher posterior probability is chosen as the prediction (e.g., since P(Yes|Sunny) > P(No|Sunny), the player can play).

**Advantages of Naïve Bayes Classifier** include:

- **Speed and Simplicity:** It is one of the **fast and easy** machine learning algorithms for predicting a class.
- **Versatility:** It can be applied to both **Binary and Multi-class Classifications**.
- **Performance in Multi-class:** It often performs well in **Multi-class predictions** compared to other algorithms.

- **Text Classification:** It is a popular choice for **text classification problems**, such as spam filtering and sentiment analysis.
- **Real-time Predictions:** Being an **eager learner**, it can be effectively used for **real-time predictions**.

The main **disadvantage** of Naïve Bayes Classifier is its **feature independence assumption**. Because it assumes all features are independent or unrelated, it **cannot learn complex relationships between features**, which may not hold true in real-world data.
**Applications of Naïve Bayes Classifier** include:

- **Credit Scoring**.
- **Medical data classification**.
- **Real-time predictions**.
- **Text classification**, such as **Spam filtering and Sentiment analysis**.

### Q13: Define quartiles and the Interquartile Range (IQR), explaining how IQR is used to detect outliers.

**Quartiles** are statistical measures that **divide a dataset into four equal parts**, with each quartile representing **25% of the data**. The three main quartiles are:

1. **Q1 (First Quartile):** This is the **25th percentile**, meaning 25% of the data points fall below this value. It is also known as the **lower quartile**.
2. **Q2 (Second Quartile):** This is the **50th percentile**, which is also the **median** of the dataset, indicating that 50% of the data points are below this value.
3. **Q3 (Third Quartile):** This is the **75th percentile**, meaning 75% of the data points are below this value. It is also known as the **upper quartile**.

The **Interquartile Range (IQR)** is defined as the range between the **third quartile (Q3)** and the **first quartile (Q1)**. It is calculated using the formula: **IQR = Q3 - Q1**. The IQR measures the **spread of the middle 50% of the data** and is considered **robust to outliers** because it focuses on this central portion of the data.
The **1.5 × IQR rule** is a commonly used method to **detect outliers**. A data point is considered an outlier if it falls outside the following boundaries:

1. A value is considered a **low outlier** if it is **below Q1 - 1.5 × IQR**.
2. A value is considered a **high outlier** if it is **above Q3 + 1.5 × IQR**.

### Q14: Explain Logistic Regression, covering its purpose, key concepts, steps involved, and its common applications.

**Logistic Regression** is a **statistical method** specifically designed for **binary classification problems**. Unlike linear regression, which predicts continuous values, logistic regression **predicts the probabilities** that an instance belongs to one of two possible classes (typically represented as 0 or 1).
Its **purpose** is to predict the probability of a **binary outcome**, such as yes/no, pass/fail, or spam/not spam. It is used when the outcome variable can only take one of two possible values or classes.

**Key Concepts** in Logistic Regression include:

1. **Binary Outcome:** The outcome variable must be binary, meaning it can only take one of two values.
2. **Logistic (Sigmoid) Function:** This function is central to logistic regression. It **maps the predicted values to probabilities between 0 and 1**, ensuring that the model's output is always within a valid probability range. The sigmoid function ensures that

regardless of the input value, the output probability is always constrained within the valid range.

3. **Decision Boundary:** After computing a probability, a **threshold (commonly 0.5)** is applied. If the predicted probability exceeds this threshold (e.g., > 0.5), the instance is classified as the positive class (1); otherwise, it is classified as the negative class (0).
4. **Odds & Log-Odds:** Logistic regression models the relationship between the features and the **log-odds of the target variable**, making it a linear model in the log-odds space.

The **steps involved in Logistic Regression** are:

1. **Training:** The model learns the optimal parameters (beta coefficients) for the dataset, typically using **maximum likelihood estimation (MLE)**.
2. **Prediction:** After training, the model can predict probabilities for new data. These probabilities are then converted into binary outcomes (0 or 1) based on the chosen decision threshold.

A key takeaway is that the output provides not only a binary prediction but also a probability, offering insight into the model's confidence.

**Applications of Logistic Regression** are widespread and include:

- **Medicine:** For predicting the presence of diseases.
- **Finance:** In credit scoring.
- **Marketing:** For predicting customer churn.

### Q15: Describe the Support Vector Machine (SVM) algorithm, its main goal, how it works, including its handling of non-linearly separable data, and its properties and issues.

**Support Vector Machine (SVM)** is a popular **supervised learning algorithm** used for both **classification and regression problems**.

The **main goal of the SVM algorithm** is to **create the best line or decision boundary (hyperplane)** that can segregate n-dimensional space into different classes. This ensures that new data points can be easily and accurately assigned to the correct category in the future. SVM achieves this by choosing **extreme points/vectors, known as support vectors**, to define this hyperplane.

**How SVM Works** involves several key concepts:

1. **Hyperplane:** This is the **decision boundary** used to separate data points of different classes in a feature space. For linear classifications, it is represented by a linear equation (e.g., $wx + b = 0$).
2. **Maximum Margin (Hard Margin):** SVM aims to find a hyperplane that creates the **largest possible separation or margin** between the two classes. This hyperplane is positioned to maximize its distance to the nearest data point from each class.
3. **Handling Outliers (Soft Margin):** SVM can tolerate some misclassifications or outliers by using **soft margins**. If data points fall within or cross the margin, SVM adds a penalty for each violation, attempting to minimize (1/margin + ∑penalty), where hinge loss is a common penalty. This makes SVM robust to outliers.
4. **Non-linearly Separable Data (Kernel Trick):** When data cannot be linearly separated in its original feature space, SVM uses a **kernel function** (the "kernel trick"). A kernel function is a non-linear mathematical function that **transforms the input data into a higher-dimensional feature space** where the data points may become linearly separable. SVM then finds a linear hyperplane in this new higher-dimensional space. Common kernel functions include the **linear kernel, polynomial kernel, and Gaussian kernel**.

- **Types of SVM** include:
    - **Linear SVM:** Used for data that can be classified into two classes by a single straight line (linearly separable data).
    - **Non-linear SVM:** Used for data that cannot be separated by a straight line (non-linearly separated data).
- **Properties of SVM** include:
    - **Flexibility** in choosing a similarity function.
    - **Sparseness of solution:** Only support vectors are used to define the separating hyperplane, making it efficient for large datasets.
    - Ability to handle **large feature spaces**; complexity doesn't depend on dimensionality.
    - **Overfitting can be controlled** by the soft margin approach.
    - Involves a **simple convex optimization problem** guaranteed to converge to a single global solution.
    - Allows for **Feature Selection**.
- **Issues (Disadvantages) in SVM** are:
    - Not suitable for **large datasets**.
    - **Does not perform well** when the dataset has **more noise** (i.e., target classes are overlapping).
    - **Underperforms** when the number of features for each data point **exceeds the number of training data samples**.

### Q16: Explain the Confusion Matrix, defining its components and listing four key metrics derived from it.

A **confusion matrix** is a crucial **performance measurement tool** for classification algorithms in machine learning. Its primary **purpose** is to help **visualise the performance** of an algorithm by showing the counts of both correct and incorrect predictions. It is structured as a **square matrix** where the **rows represent the actual classes** and the **columns represent the predicted classes**.

The **components of a Confusion Matrix** are essential for understanding model performance:

1. **True Positive (TP):** Instances that were **actually positive and were correctly predicted as positive**.
2. **True Negative (TN):** Instances that were **actually negative and were correctly predicted as negative**.
3. **False Positive (FP):** Instances that were **actually negative but were incorrectly predicted as positive** (also known as Type I Error).
4. **False Negative (FN):** Instances that were **actually positive but were incorrectly predicted as negative** (also known as Type II Error).

The typical structure is:

| | **Predicted Positive** | **Predicted Negative** |
|---|---|---|
| **Actual Positive** | TP | FN |
| **Actual Negative** | FP | TN |

**Four key metrics derived from the Confusion Matrix** are:

1. **Accuracy:** Calculated as **(TP+TN) / (TP+FP+FN+TN)**. It measures the **overall correctness** of the model's predictions.
2. **Precision:** Calculated as **TP / (TP+FP)**. It indicates the **proportion of true positives among all instances predicted as positive**.

3. **Recall (Sensitivity):** Calculated as **TP / (TP+FN)**. It measures the model's ability to **identify all actual positive instances**.
4. **F1-Score:** Calculated as **2 × (Precision × Recall) / (Precision + Recall)**. This is the **harmonic mean of precision and recall**, and it is particularly useful for **imbalanced datasets**.

### Q17: Describe the K-Nearest Neighbors (KNN) algorithm, its working principle, and list its advantages, limitations, and key characteristic regarding learning.

**K-Nearest Neighbors (KNN)** is a **simple, instance-based machine learning algorithm** used for both **classification and regression tasks**.
Its **working principle** involves classifying new data points based on the majority class of their 'k' nearest neighbors in the training dataset. The specific steps are:

1. **Choose 'k':** Determine the number of nearest neighbors to consider (e.g., k=3). The choice of 'k' significantly influences the model's performance.
2. **Calculate Distances:** Measure the distance (commonly Euclidean distance) between the new data point and every existing point in the training dataset.
3. **Identify Nearest Neighbors:** Select the 'k' closest data points based on these calculated distances.
4. **Assign Majority Class:** The new data point is then assigned to the class that is most frequent among its 'k' nearest neighbors. For example, if three nearest neighbors belong to classes A, A, and B, the new point is classified as A.

- **Advantages of KNN** include:
  - **Simplicity and Ease of Implementation:** It does not involve a complex training phase, making it straightforward to use.
  - **Adaptability:** It works effectively with **multi-class classification problems**.
- **Limitations of KNN** are:
  - **Computationally Intensive:** It can be **slow for large datasets** because it needs to calculate distances to all training points during the prediction phase.
  - **Sensitive to Noise and Outliers:** Its performance can be negatively impacted by irrelevant or noisy features.

A key characteristic of KNN regarding learning is its nature as a **lazy learning algorithm**. This means that KNN **makes decisions only during prediction time**, rather than learning an explicit model during a dedicated training phase.